[COMPANY NAME]

TECHNICAL PRODUCT MEMORANDUM

[PRODUCT NAME]

Codename: VALKYRJA

INTELLIGENCE-GRADE MOBILE COMPUTING PLATFORM

Document Type	Technical Product Memorandum — Manufacturer & Supplier Brief
Document Version	1.0 — Initial Release
Prepared By	Desirae Stark, Chief Architect
Organization	[COMPANY NAME]
Date	April 2026
Classification	PROPRIETARY & CONFIDENTIAL
Target Audience	Hardware Manufacturers, Component Suppliers, Software Integration Teams

PROPRIETARY NOTICE

This document contains proprietary technical specifications, architectural designs, and trade secrets belonging to [COMPANY NAME]. It is disclosed solely for the purpose of manufacturing, supply, and integration negotiations. Unauthorized disclosure, reproduction, or use is strictly prohibited.
All specifications subject to revision. Recipients must execute a Non-Disclosure Agreement prior to receipt. Intellectual property rights reserved. Patent applications pending.

SECTION — EXECUTIVE SUMMARY

Product Vision & Strategic Context

Purpose. This memorandum constitutes the authoritative technical specification for [PRODUCT NAME] (Codename: VALKYRJA), a purpose-engineered mobile computing platform developed by [COMPANY NAME] under the QWAMOS operating system architecture. It is intended to provide hardware manufacturers, component suppliers, and software integration partners with the complete specification required to procure, fabricate, assemble, and validate the device.

Mission. VALKYRJA is the world's first commercial mobile device to deliver intelligence-grade operational security as the default user experience — not as a configuration layer, not as a third-party add-on, but as the foundational architecture from which all other capabilities are derived. The device targets a market gap that no existing flagship addresses: a phone that simultaneously exceeds all commercial flagships in raw computational performance, matches their user experience quality, and provides security capabilities previously available only through classified government platforms.

Differentiation. Every current 'secure phone' sacrifices performance or usability to achieve security. Every current flagship sacrifices security to achieve performance and UX. VALKYRJA is architected to deliver all three simultaneously through a layered approach: hardware isolation enforced at silicon level, post-quantum cryptography throughout the full stack, a KVM hypervisor providing VM-level compartmentalization, and a consumer-grade UX abstraction layer that makes the complexity invisible to the end user.

Operating System. VALKYRJA ships exclusively with QWAMOS v3.1.0+ (Qubes+Whonix Advanced Mobile Operating System), an open-source (AGPL-3.0) ARM64-native hypervisor OS with 27 completed development phases, 50,000+ lines of code, and full post-quantum cryptographic hardening throughout the stack. QWAMOS software specifications are detailed in Section 10 of this memorandum.

Market Tiers

Tier	Designation	Target Market
Tier 1	Consumer Privacy	Privacy-conscious consumers, journalists, executives requiring corporate data separation. Volume production device.
Tier 2	Professional Security	Security researchers, legal professionals, enterprise mobile security, financial institutions, diplomatic corps.
Tier 3	Operational	Intelligence community adjacent, government contractors, cleared personnel. Custom configuration per deployment.

Performance Targets vs. Market Comparators

Capability	VALKYRJA	Samsung Galaxy S26 Ultra	Google Pixel 10 Pro XL
Hypervisor OS	Native KVM — full Type-1	None	AVF/pKVM (limited)
RAM	32 GB LPDDR5X	12–16 GB LPDDR5X	16 GB LPDDR5X
Internal Storage	1 TB UFS 4.0 + 512 GB UFS 4.0	256 GB – 1 TB	128 GB – 1 TB
Privacy Display	FMP LEAD 2.0 + FLAG_SECURE (dual enforcement)	FMP LEAD 2.0 (software only)	None

Network Anonymization	Mandatory Tor/I2P all traffic	None	None
PQC Cryptography	Full stack — all comms, storage, keys	None	Partial
Hardware Kill Switches	4× physical relay	None	None
Security Microcontroller	Discrete HSM + nuclear-powered HNCP	TrustZone only	Titan M2
VM Isolation	8 isolated domains	App sandbox	App sandbox
Charge Time (0→100%)	6–10 min (graphene cell)	~80 min	~70 min

SECTION 1 COMPUTE & PROCESSING ARCHITECTURE

Primary SoC, GPU, NPU, and Supporting Silicon

1.1 Primary System-on-Chip: Rockchip RK3588

The primary compute engine is the Rockchip RK3588, selected as the sole ARM64 SoC in the market that satisfies the non-negotiable requirement of full EL2 (Exception Level 2) hypervisor access, enabling native KVM virtualization. This property — absent from all Qualcomm Snapdragon, Samsung Exynos, and MediaTek Dimensity SoCs due to proprietary hypervisor firmware occupying EL2 — is fundamental to QWAMOS's security architecture and cannot be substituted.

PARAMETER	SPECIFICATION
Designation	Rockchip RK3588
Process Node	8nm FinFET (Samsung Foundry)
CPU Cluster A (Performance)	4× ARM Cortex-A76 @ 2.4 GHz
CPU Cluster B (Efficiency)	4× ARM Cortex-A55 @ 1.8 GHz
CPU Architecture	ARM DynamIQ big.LITTLE, ARMv8.2-A ISA
Total CPU Cores	8 (Octa-core)
L2 Cache (A76)	1 MB per core (4 MB total)
L2 Cache (A55)	512 KB per core (2 MB total)
L3 Cache	3 MB shared
GPU	ARM Mali-G610 MP4 (4-core), OpenGL ES 3.2, Vulkan 1.2, OpenCL 2.2
GPU Clock	Up to 1,000 MHz
NPU	6 TOPS (Tera Operations Per Second), 3-core neural processor
VPU (Video)	8K@60fps decode; 8K@30fps encode; AV1, H.265, H.264, VP9
ISP	Dual-core ISP, up to 48 MP + 48 MP simultaneous
Memory Interface	64-bit LPDDR4/LPDDR5, dual-channel
Max Memory Bandwidth	68.3 GB/s (LPDDR5 @ 4266 MHz)
PCIe	2× PCIe 3.0 x4 + 1× PCIe 2.0 x1
USB	USB 3.1 Gen1 + USB 2.0 OTG
Display Interface	2× MIPI DSI (4-lane each), 1× eDP
EL2 / KVM	CONFIRMED — kernel boots at EL2, /dev/kvm accessible
TDP	~10–14 W peak (SBC reference); ~4–8 W mobile thermal envelope
Thermal Design	ARM DynamIQ DVFS, hardware thermal throttling, software governer

1.2 CPU Architecture & QWAMOS VM Allocation

The 8-core DynamIQ cluster is allocated across QWAMOS's virtual machine domains as follows. This allocation is enforced by the AI Governor module (hypervisor/ai_governor.py) with real-time workload prediction and dynamic rebalancing.

VM Domain	CPU Allocation	Priority	Notes
Dom0 / Control VM	1× A76 + 2× A55 (reserved)	Highest — real-time	ML threat detection, VM lifecycle, security policies
Gateway VM (Tor/I2P)	1× A76 + 1× A55 (dedicated)	High — sustained	Always-on; Tor circuits, I2P, DNSCrypt, behavioral obfuscation
Active User VM	2× A76 + 1× A55 (primary burst)	High — interactive	Android VM, Arch VM, or Kali VM — whichever profile is active
Background VMs	Remaining A55 cores	Low — background	Suspended or minimal-activity domains
AI Governor	NPU (6 TOPS) — dedicated	Asynchronous	ML resource scheduling, threat detection inference, LLM inference
GPU Isolation	Mali-G610 MP4 — per-VM sliced	As-needed	Vulkan passthrough per VM; no cross-VM GPU access

1.3 Supporting Security Silicon

Beyond the primary SoC, VALKYRJA incorporates discrete security silicon operating independently of the main compute domain. These components are non-negotiable and must be sourced and integrated in parallel with primary SoC procurement.

COMPONENT	SPECIFICATION
Discrete HSM	Infineon SLB 9672 TPM 2.0 or equivalent — dedicated post-quantum key management, attestation, ML-DSA-87 operations. Communicates via SPI to RK3588S; key material never exported in plaintext.
Hardware TRNG	Dedicated true random number generator IC — ring oscillator entropy + reverse-biased junction shot noise. Seeds HSM, KVM hypervisor entropy pool, and all VM-level key generation. Independent of SoC PRNG.
HNCP (Hardware Network Co-Processor)	Ultra-low-power RISC-V microcontroller (e.g., SiFive FE310 class, <40 µW). Sits physically on-wire between cellular modem and RK3588. Enforces traffic isolation rules in hardware. Powered by dedicated nuclear security rail — never from primary battery. Cannot be overridden by software.
Tamper Detection MCU	ARM Cortex-M0+ (STM32L0 class, <1 µW stop mode). Monitors physical tamper conditions: accelerometer, case-open sensor, light sensor, temperature anomaly. Powered by nuclear security rail. Triggers key destruction on tamper confirmation, even with primary battery removed.
Biometric Enclave	Discrete ARM SecurCore SC300 — dedicated to fingerprint template storage and matching. Isolated from main SoC; communicates result only (match/no-match) over hardened interface. Under-display optical fingerprint sensor integration.
Secure Boot Chain	Custom U-Boot bootloader with verified boot (AVB2.0), hardware root of trust anchored in HSM. Kernel signature verified before EL2 handoff. Custom AVB keys allow bootloader re-lock with QWAMOS, maintaining verified boot integrity under custom OS.

KVM/EL2 REQUIREMENT — NON-SUBSTITUTABLE

The RK3588 is specified due to its confirmed EL2 access for native KVM. NO substitution with Qualcomm Snapdragon,
Samsung Exynos, or MediaTek Dimensity SoCs is acceptable. These SoCs occupy EL2 with proprietary

hypervisors
(QHEE/HypX/Gunyah) that cannot be removed, making QWAMOS's KVM-based VM isolation architecturally impossible.
Any proposed SoC substitution must provide documented EL2 access confirmation before evaluation.

SECTION 2 MEMORY ARCHITECTURE

Maximum RAM Configuration & Allocation

2.1 Primary RAM — LPDDR5X

VALKYRJA is specified at the maximum RAM configuration to support simultaneous operation of multiple KVM virtual machines, ML threat detection models, behavioral obfuscation inference, and the full QWAMOS security stack without memory pressure-induced performance degradation.

PARAMETER	SPECIFICATION
Total Capacity	32 GB LPDDR5X (maximum supported by RK3588 memory controller)
Standard	LPDDR5X — JEDEC JESD209-5B
Data Rate	8,533 Mbps per pin
Interface Width	64-bit dual-channel
Peak Bandwidth	68.3 GB/s
Package	PoP (Package-on-Package) or discrete FBGA — integrated with SoC
Operating Voltage	0.5V (VDDQ) — ultra-low power LPDDR5X spec
ECC	Optional inline ECC — recommended for Dom0 / HSM communication channels
Temperature Range	0°C to 85°C (commercial); -20°C to 85°C (extended industrial option)
Preferred Suppliers	Samsung Semiconductor (K4X), SK Hynix (H58G), Micron (MT62F)

2.2 VM Memory Allocation Budget

The following table defines the baseline memory allocation per VM domain at system initialization. The AI Governor dynamically adjusts these allocations based on active workload. Total static allocation of 26 GB leaves 6 GB as dynamic pool for burst VM operations, snapshot buffers, and ML model loading.

VM Domain	Base Allocation	Max Burst	Purpose
Dom0 / Control VM	2 GB	4 GB	Hypervisor management, ML detection models (Autoencoder, RF, LSTM), security policies
Gateway VM	2 GB	3 GB	Tor daemon, I2P daemon, DNSCrypt, VPN, WebTunnel, V2Ray, behavioral obfuscation ML
Android VM (Primary)	6 GB	10 GB	Primary user Android environment, app ecosystem, daily use
Secure VM (Arch Linux)	4 GB	6 GB	PQC communications, Signal+ZRTP, steganography, secure file operations
Kali NetHunter VM	3 GB	5 GB	Offensive security, penetration testing, vulnerability assessment
Vault VM (Air-gapped)	1 GB	2 GB	Key storage, cryptographic operations — no network interface
Ubuntu Dev VM	2 GB	4 GB	Development environment, AI App Builder pipeline

Disposable VM	2 GB	4 GB	Ephemeral — fresh instantiation per session, auto-wipe on close
Kernel / Hypervisor overhead	2 GB	2 GB	KVM hypervisor, VirtIO drivers, NIC enforcer, GPU manager
Dynamic Pool	6 GB	6 GB	Burst buffer, snapshot staging, model loading, ML inference staging
TOTAL	32 GB	—	Full platform allocation — no swap partition; all operations in encrypted RAM

RAM ENCRYPTION REQUIREMENT

All 32 GB must operate under in-memory encryption enforced by the RK3588 memory controller in conjunction with TrustZone. No VM guest memory is accessible to Dom0 in plaintext under pKVM protected mode. Secure VMs operate with hardware-enforced memory isolation — physical memory pages cannot be read by any other domain, including the hypervisor itself. This is architecturally enforced, not policy-enforced.

SECTION 3 STORAGE ARCHITECTURE

Maximum Capacity, UFS 4.0, Encrypted Volume Stack

3.1 Primary Storage — UFS 4.0

PARAMETER	SPECIFICATION
Capacity	1 TB UFS 4.0 (primary)
Standard	JEDEC UFS 4.0 (JESD220F)
Sequential Read	Up to 4,200 MB/s
Sequential Write	Up to 2,800 MB/s
Random Read (4K)	Up to 100,000 IOPS
Random Write (4K)	Up to 70,000 IOPS
Interface	UniPro / M-PHY Gen5 — PCIe 4.0 equivalent bandwidth
Form Factor	BGA embedded (eMCP consideration for board space)
Encryption	Hardware AES-256 inline encryption at UFS controller level (baseline layer)
Preferred Suppliers	Samsung Semiconductor (KLUDG), SK Hynix (H28U1), Micron (MTFC)

3.2 Secondary Storage — PCIe NVMe

A second storage domain is provided via an internal PCIe 3.0 x4 M.2 slot (custom form factor for mobile integration), enabling a dedicated encrypted storage volume for classified material, large VM images, and forensic evidence chain-of-custody logs separate from the primary UFS volume.

PARAMETER	SPECIFICATION
Capacity	512 GB NVMe SSD (secondary — PCIe 3.0 x4)
Interface	PCIe 3.0 x4 via RK3588 PCIe controller
Sequential Read	Up to 3,500 MB/s
Sequential Write	Up to 3,000 MB/s
Form Factor	Custom M.2 2230 or embedded BGA NVMe
Assignment	Dedicated to Vault VM encrypted volume — air-gapped from network VMs
Encryption	QWAMOS PQC Volume (ML-KEM-1024 + ChaCha20-Poly1305) on top of hardware AES

3.3 Storage Allocation Map

VOLUME	CAPACITY & ASSIGNMENT
QWAMOS Base System (UFS)	64 GB — Hypervisor, Dom0, Gateway VM, kernel, U-Boot
Android VM Image (UFS)	200 GB — Primary Android environment, apps, media
Secure VM Image (UFS)	100 GB — Arch Linux secure comms environment
Kali NetHunter VM (UFS)	100 GB — Offensive security tools and datasets
Ubuntu Dev VM (UFS)	100 GB — Development environment and project files

Disposable VM Pool (UFS)	50 GB — Ephemeral VM instantiation pool (overwritten each session)
VM Snapshot Buffer (UFS)	150 GB — Rollback snapshots for all active VMs
Encrypted User Data (UFS)	236 GB — General encrypted user storage, accessible per profile
Vault VM Volume (NVMe)	512 GB — Air-gapped classified storage, key material, evidence logs
TOTAL USABLE	~1.5 TB across UFS + NVMe

3.4 Encryption Stack

All storage volumes are encrypted under a layered architecture. Breaking any single encryption layer does not compromise data. The innermost layer (hardware AES in UFS/NVMe controller) provides base-layer protection. QWAMOS's post-quantum volume encryption provides the outer protection.

Layer	Algorithm	Scope
Layer 1 (Hardware)	AES-256 XTS — UFS/NVMe hardware inline	All physical storage
Layer 2 (QWAMOS PQC Volume)	ML-KEM-1024 (key encapsulation) + ChaCha20-Poly1305 (AEAD)	Per-VM encrypted volumes
Layer 3 (Key Derivation)	Argon2id — GPU/ASIC-resistant KDF from passphrase	Master key derivation
Layer 4 (Key Storage)	ML-DSA-87 — hardware-backed in Infineon HSM	Key material protection
Secure Deletion	3-pass DoD-grade wipe + AES key destruction	On panic wipe / VM teardown

SECTION 4 DISPLAY ARCHITECTURE

Samsung LEAD 2.0 FMP — Privacy Display Integration

4.1 Panel Specification

PARAMETER	SPECIFICATION
Panel Technology	Samsung Display LEAD 2.0 AMOLED with Flex Magic Pixel (FMP) — privacy display
Privacy Technology	Flex Magic Pixel — dual pixel type (Narrow + Wide) with multi-layer Black Matrix architecture
Diagonal Size	6.9 inches
Resolution	3088 × 1440 px (WQHD+)
Pixel Density	~490 PPI
Refresh Rate	1–120 Hz LTPO 4.0 adaptive
Peak Brightness (Standard)	2,600 nits
Brightness (Privacy Mode)	Compensated to maintain front-facing legibility; narrow pixel luminance boosted
Privacy Mode (45° off-axis)	3.5% front brightness retained (vs. ~40% on conventional OLED)
Privacy Mode (60° off-axis)	0.9% or less front brightness retained — screen appears black
Color Gamut	DCI-P3 > 99%, sRGB 100%
Color Depth	16 million colors, 10-bit depth
HDR	HDR10+ certified
Under-display sensor	Optical fingerprint sensor — routed to discrete Biometric Enclave (not SoC)
FLAG_SECURE	Enforced at Flutter compositor level across all VM domains — no screen capture possible
Touch	Capacitive multi-touch, 10-point, integrated in panel
Supplier	Samsung Display Corporation — LEAD 2.0 / FMP panel; direct B2B engagement required

4.2 FMP Privacy Display — Technical Detail

Flex Magic Pixel operates through two distinct pixel types interlaced across the display substrate. Wide-emitting pixels disperse light across a broad angular cone (standard viewing). Narrow-emitting pixels direct light forward within a tight cone (~30° half-angle). When Privacy Mode is active, wide pixels are deactivated or driven to near-zero luminance; only narrow pixels emit. The Black Matrix structure — modified to a multi-layer configuration by Samsung Display — further collimates light from narrow pixels, amplifying the angular restriction effect.

Mode	Pixel State	Side Visibility
Off (Standard)	Wide + Narrow both active	~40% brightness at 45° — normal smartphone behavior
Privacy Mode	Narrow active; Wide minimized	3.5% brightness at 45°; 0.9% at 60°
Maximum Privacy	Narrow active; Wide at mid-gray fill	Near-zero at 60°; dark gray blocks to side viewers

4.3 QWAMOS Profile-Native FMP Integration

VALKYRJA's implementation exceeds Samsung's own S26 Ultra integration in a critical respect: FMP mode is driven by QWAMOS profile state from Dom0, not by a user toggle or application-level API. The display controller receives a hardware signal directly from the hypervisor control plane, bypassing the user VM software stack entirely. No application running inside any VM can override the display privacy mode for its profile tier.

Profile	FMP State	FLAG_SECURE	Rationale
Everyday	Off (wide emission)	Active (all screens)	Normal use; shoulder surfing less probable; UX priority
Private	Standard Privacy (30° cone)	Active	Moderate threat; transaction protection, messaging
Secure	Maximum Privacy (tight cone)	Active	High threat; classified comms; all sensitive ops
Ghost	Maximum Privacy (tight cone)	Active	Hostile environment; disposable VM; full countersurveillance
Duress	Off (standard)	Active	Decoy profile must appear normal to coercive observer

4.4 Secondary Shoulder-Surf Defense Layers

FMP alone addresses optical shoulder-surfing. VALKYRJA combines it with two additional independent defenses:

- SecureType Keyboard: every keystroke encrypted at hardware layer before leaving keyboard subsystem, eliminating value of shoulder-surfing keystrokes even if FMP is bypassed
- Keyboard layout randomization: layout reshuffled every 30 seconds, defeating layout memorization attempts
- FLAG_SECURE at Flutter compositor: eliminates software-layer screen capture across all VM domains regardless of FMP state

DISPLAY SUPPLIER PROCUREMENT NOTE

Samsung Display Corporation supplies FMP/LEAD 2.0 panels to third-party device manufacturers as a commercial
panel component (separate from Samsung Electronics device business). Samsung Display actively promoted LEAD 2.0
to global customers at MWC 2026, and independent UL Solutions certification facilitates third-party licensing.
Initial B2B engagement should target Samsung Display's Mobile Display Business Division.
Minimum order quantity negotiation is the primary commercial challenge; Tier 1 consumer volume commitment
enables Tier 2/3 operational device supply on the same panel production line.

SECTION 5 POWER ARCHITECTURE

Four-Domain Independent Power System

5.1 Architecture Overview

VALKYRJA employs a four-domain power architecture in which each domain is electrically isolated at the PMIC level with no cross-domain power flow paths. This isolation ensures that an adversary who drains, removes, or compromises any single power domain cannot affect the security properties governed by other domains.

Domain	Source	Powers
Domain 1 — Solar Harvesting	Perovskite QD conformal panel (rear)	Feeds Domain 2 primary cell via isolated PMIC charge path
Domain 2 — Primary Compute Cell	Graphene-silicon Li-ion + QD electrode enhancement	RK3588, all VMs, display, 5G modem, WiFi, storage, all compute
Domain 3 — Nuclear Security Rail	Betavolt BV100 betavoltaic cell	HNCP, Tamper MCU, RTC — permanently, without interruption
Domain 4 — Physical Kill Circuits	Passive — no power required	Modem RF, microphone ground short, camera power, GPS power

5.2 Primary Cell — Graphene-Silicon Li-Ion

PARAMETER	SPECIFICATION
Cell Chemistry	Graphene-silicon composite anode Li-ion (graphene-stabilized silicon anode)
Nominal Capacity	7,700 mAh effective (nominal 5,500 mAh graphene-silicon, +40% vs. standard Li-ion)
Electrode Enhancement	Graphene quantum dot (GQD) electrode additives — additional ~20% effective energy density
Effective Energy	~35 Wh total usable energy
Nominal Voltage	3.85 V
Charge Protocol	USB Power Delivery 3.1 — up to 100W (28V/5A profile)
Charge Time (0→100%)	6–10 minutes at full 100W PD input
Cycle Life	2,000+ cycles to >80% capacity (graphene-silicon vs. ~500 cycles standard Li-ion)
Thermal Range	-20°C to +60°C operational; no thermal runaway due to graphene anode
Cell Format	Custom pouch cell — form-fit to chassis dimensions, designed for device-specific PCB clearance
Procurement	Custom cell engagement required: Lyten (US), GMG (Australia), or equivalent graphene-silicon supplier
Certification	IEC 62133, UN 38.3, UL 2054 — required for all tier configurations

5.3 Ambient Harvesting — Quantum Dot Photovoltaics

PARAMETER	SPECIFICATION
Technology	Perovskite quantum dot solar cell — conformal flexible film

Placement	Rear panel — full-surface conformal lamination beneath protective coating
Efficiency	18.2% PCE (perovskite QD, current research frontier)
Output (bright indoor)	100–150 mW
Output (office ambient)	50–80 mW
Output (outdoor, direct)	200–400 mW (depending on panel area)
Role	Trickle harvesting — extends field operation time 15–25%; does NOT replace primary cell
Charge Path	Isolated MPPT controller → Domain 2 primary cell; no path to Domain 3 nuclear rail
Procurement	Custom conformal deposition — Saule Technologies, Quantum Solutions, or equivalent

5.4 Nuclear Security Rail — Betavoltaic Cell

PARAMETER	SPECIFICATION
Technology	Betavoltaic — diamond semiconductor with Nickel-63 isotope
Product Reference	Betavolt BV100 (or equivalent)
Output Power	100 µW continuous
Output Voltage	3V
Dimensions	15 × 15 × 5 mm
Operational Life	50 years (Ni-63 half-life: 100 years; BV100 specified at 50-year operational life)
Decay Product	Ni-63 → Cu-63 (stable, non-radioactive — no environmental hazard)
External Radiation	None (diamond matrix provides complete beta shielding)
Temperature Range	-60°C to +120°C
Powers	HNCP (20–40 µW) + Tamper MCU (<1 µW) + RTC (<0.1 µW) + kill switch circuit (5 µW)
Rail Isolation	No electrical connection to Domain 2 primary cell or PMIC — completely isolated ground plane
Regulatory	NRC (US) or national equivalent licensing required for radioactive isotope incorporation
Procurement	Betavolt New Energy Technology (Beijing) — direct commercial engagement; regulatory pre-clearance required

5.5 Operational Runtime Estimates

Usage Scenario	Active Systems	Net Draw	Runtime
Deep Standby	Dom0 + Gateway VM only; screen off; radios idle	~1.2 W	28–32 hours
Tactical Standby	Full QWAMOS stack; 5G connected; periodic Tor comms	~2.1 W	14–17 hours
Light Daily Use	Screen on; 1 active VM; WiFi;	~3.7 W	8–10 hours

	browsing / messaging		
Moderate Operational Use	Screen on; 2–3 VMs active; WiFi + 5G; comms active	~9.1 W	3.5–4.5 hours
Heavy Use	All VMs; screen max brightness; 5G data; crypto ops	~16.8 W	1.8–2.5 hours
Solar Supplement (outdoor)	Any scenario above — adds 200–400 mW ambient	Net reduction	+15–25% runtime
Recharge (0 → 100%)	USB-C PD 100W input	—	6–10 minutes

SECTION 6 HARDWARE SECURITY ARCHITECTURE

Silicon-Level Security — Below Software Attack Surface

6.1 Security Domain Isolation

VALKYRJA's security architecture enforces isolation at four distinct layers, each independent of the others. Compromise of any single layer does not cascade to others. The security properties are enforced progressively from hardware (cannot be overridden by software) upward through firmware, hypervisor, and OS.

Security Layer	Mechanism	Attack Vectors Addressed
Layer 0 — Physics	Nuclear-powered HNCP on-wire; physical kill switch circuits	Battery removal defeat; modem bypass; physical coercion
Layer 1 — Silicon	Discrete HSM, hardware TRNG, Biometric Enclave, TrustZone	Key extraction, side-channel, bootloader attacks
Layer 2 — Firmware	Verified U-Boot, custom AVB keys, ML bootloader integrity verification	Evil maid, firmware persistence (Vault 7 style), cold boot
Layer 3 — Hypervisor	KVM at EL2; pKVM memory isolation; VM domain separation	Cross-VM lateral movement, privilege escalation
Layer 4 — OS/Software	QWAMOS RASP, ML threat detection, nftables firewall, Tor routing	Zero-click exploits, network surveillance, data exfiltration

6.2 Hardware Kill Switches

Four recessed physical kill switches are located on the right spine of the device. These open physical electrical circuits — not GPIO lines controlled by software. No firmware, no OS, no hypervisor can restore connectivity once a kill switch is engaged. Each switch is a miniature normally-closed relay actuated by the physical switch position.

SWITCH	CIRCUIT ACTION
[1] NETWORK	Opens RF power supply circuit to cellular modem module. The modem loses power entirely — no 4G/5G/WiFi/Bluetooth transmission possible. Independent of software radio stack.
[2] MICROPHONE	Shorts microphone output lines to ground potential. No audio signal can propagate regardless of software state. Hardware-level acoustic isolation.
[3] CAMERA	Disconnects power rail to all camera modules (front, rear triple-lens system). Sensor draws no power; no image capture possible.
[4] LOCATION	Disconnects power to GPS receiver and interrupts location data lines from cellular modem. Software location APIs return null or last-known value.

6.3 Emergency Protection Systems

SYSTEM	TRIGGER & MECHANISM
Panic Wipe	5-finger triple-tap gesture (QWAMOS gesture engine, Dom0). Destroys all VM encryption keys via HSM key deletion command. UFS and NVMe AES keys overwritten with random data. Executed in <2 seconds from gesture detection.
Duress PIN	Alternate PIN entered at unlock. Boots convincing Android VM decoy profile. Primary VMs remain suspended in encrypted state, invisible to decoy environment. No forensic trace of QWAMOS hypervisor visible.
Nuclear Tamper Wipe	HNCP/Tamper MCU detects tamper condition (case opening, abnormal acceleration, temperature spike consistent with decapping). Sends hardware signal to HSM — key material destroyed at silicon level before main SoC can boot.

	Operates with zero main battery level.
VM Snapshot Rollback	Any VM can be restored from encrypted snapshot after suspected compromise. Rollback command issued from Dom0; compromised VM state is discarded; clean snapshot restored.
Tor Kill Switch	If Gateway VM detects Tor circuit failure or network route to clearnet, nftables rules block ALL outbound traffic. No fallback to unrouted connections under any failure mode.
Network Kill Switch	Software command from any profile (Quick Panel). Equivalent to physical Network kill switch — modem RF section powered down via relay signal from HNCP.

6.4 Anti-Surveillance Hardware

FEATURE	IMPLEMENTATION
Ultrasonic Detection	18–22 kHz bandpass microphone circuit (separate from primary mic) monitors for ultrasonic tracking beacons. QWAMOS ultrasonic_guard module triggers automated speaker jamming response.
Pegasus/Graphite Zero-Click Guard	security/pegasus_graphite_guard/ module: hypervisor-level memory isolation prevents zero-click exploit chains from gaining persistence. Baseband isolated in Gateway VM — modem attack surface dramatically reduced.
TLS Fingerprint Masking	JA3/JA4 TLS fingerprint mimicry — outbound TLS handshakes match major browser fingerprints to defeat surveillance fingerprinting of QWAMOS.
Behavioral Obfuscation	ML-driven typing rhythm, mouse/touch pattern, and browsing behavior randomization. Defeats behavioral fingerprinting even against ML-based surveillance systems.
Anti-Screenshot	FLAG_SECURE enforced at display controller and Flutter compositor — screenshot and recording APIs return black frame across all applications and VMs.

SECTION 7 NETWORK & COMMUNICATIONS ARCHITECTURE

Isolated Modem — Mandatory Anonymization — PQC Transport

7.1 Cellular Modem — Isolated Domain

PARAMETER	SPECIFICATION
Module	Sierra Wireless EM9190 (or equivalent 5G Sub-6 + mmWave M.2 modem)
Standard	5G NR Sub-6 GHz + mmWave; 4G LTE fallback; 3G/2G legacy
Interface to RK3588	PCIe 3.0 x1 — routed through HNCP (Hardware Network Co-Processor)
Physical Isolation	Modem accessible ONLY via HNCP gating. No direct memory-mapped access from RK3588. VM guest code cannot address modem directly.
SIM	Dual: eSIM (primary) + physical nano-SIM (secondary) with hardware kill circuit
Antenna	4× external antenna pads — positions TBD per chassis design for optimal RF performance
Vault 7 Mitigation	Baseband confined to Gateway VM. Gateway VM compromise does not yield modem persistence — modem module operates in isolated PCIe domain.

7.2 WiFi / Bluetooth / UWB

PARAMETER	SPECIFICATION
WiFi Standard	WiFi 7 (802.11be) — 2.4 GHz / 5 GHz / 6 GHz tri-band
WiFi Speed	Up to 5.8 Gbps theoretical (2×2 MIMO, 320 MHz channel, 4096-QAM)
Bluetooth	Bluetooth 5.4 — LE Audio, LC3 codec, direction finding
UWB	Ultra-Wideband (IEEE 802.15.4z) — secure ranging, device attestation
NFC	NFC Type A/B/F, ISO 14443, ISO 15693
Module Isolation	WiFi/BT/UWB module accessible via Gateway VM only — no direct user VM access. Traffic mandatory through Tor/I2P stack.

7.3 Traffic Anonymization Stack

All outbound traffic from user VMs — regardless of protocol, application, or VM domain — must traverse the Gateway VM before reaching the cellular modem or WiFi interface. There is no software or hardware path for a user VM to bypass this requirement. The HNCP enforces this constraint at silicon level.

Layer	Technology	Function
Primary Anonymization	Tor (latest stable) — 3-hop onion routing	IP anonymization, traffic encryption, identity separation
Secondary Overlay	I2P — Invisible Internet Project	Alternative routing for I2P-native services and dead drops
DNS Protection	DNSCrypt — encrypted DNS resolution	Prevents DNS leakage; all resolution inside Gateway VM
DPI Defeat	WebTunnel transport — disguises Tor as HTTPS to Cloudflare/Fastly/Akamai CDN	Defeats deep packet inspection; traffic indistinguishable from standard HTTPS

Protocol Obfuscation	V2Ray — WebSocket, HTTP/2, gRPC, QUIC support	Multi-protocol fallback; circumvents protocol-level blocking
Bridge Protocols	obfs4, meek, Snowflake pluggable transports	Circumvents Tor-specific blocking by surveillance infrastructure
Behavioral Masking	ML-driven pattern obfuscation + Phantom Activity overlay	Defeats traffic analysis; real activity masked by synthetic decoy patterns
VPN Fallback	Multi-VPN orchestration (optional, per-profile)	Additional egress routing layer for specific operational requirements
Kill Switch	Hardware + software dual kill switches	Zero clearnet fallback under any failure mode

7.4 Secure Communications Stack

CAPABILITY	IMPLEMENTATION
Encrypted Messaging	Signal Protocol (libsignal) — E2E encrypted messaging with forward secrecy
Sealed Sender	Anonymous message delivery — hides sender metadata from Signal infrastructure
Encrypted Voice	ZRTP — media stream encryption without PKI dependency; authentication via SAS
Post-Quantum Comms	All keys wrapped with ML-KEM-1024; ML-DSA-87 signatures; hybrid construction X25519 XOR Kyber-1024
Steganography	qwamos_comms/stego/ — covert message embedding in images/files for plausibly deniable transmission
Onion Dead Drop	Tor-based asynchronous mailbox — no real-time connection requirement
Disappearing Messages	Configurable auto-deletion timers; key material destroyed on deletion

SECTION 8 PHYSICAL HARDWARE DESIGN

Chassis, PCB Stack, Connectivity & Form Factor

8.1 Form Factor

PARAMETER	SPECIFICATION
Form Factor	Candybar smartphone — single-piece aluminum alloy chassis
Display Size	6.9 inches (diagonal)
Target Dimensions	~163 × 78 × 9.2 mm (HWD) — subject to thermal and PCB stack validation
Target Weight	~245–265 g (battery + chassis + PCB stack)
Chassis Material	6000-series aluminum alloy — CNC-machined unibody; IP68 water/dust resistance
Rear Panel	Matte glass (Gorilla Glass 7i or equivalent) — conformal QD solar film laminated beneath protective coating
Finish Options	Matte Black (Tier 1); Titanium Gray (Tier 2/3 operational)
Spine — Right	4× recessed physical kill switches (Network, Microphone, Camera, Location)
Spine — Left	Volume up/down; SIM tray (nano-SIM + eSIM access)
Top Edge	USB-C 3.1 (power + data + DisplayPort Alt Mode); 3.5mm audio jack (optional)
Bottom Edge	Speaker; secondary microphone; lanyard point (Tier 3 option)

8.2 PCB Layer Stack

PCB LAYER	COMPONENTS
Layer 1 — Top (Display Assembly)	Samsung LEAD 2.0 FMP OLED panel; under-display optical fingerprint sensor; ambient light / proximity sensor
Layer 2 — Upper Main Board	RK3588 SoC (PoP with LPDDR5X stack); UFS 4.0 1 TB storage; Infineon HSM; Hardware TRNG; Biometric Enclave IC; WiFi 7 / BT 5.4 / UWB module; NFC controller
Layer 3 — Network Enforcement	HNCP RISC-V MCU; Tamper Detection MCU (Cortex-M0+); Sierra Wireless EM9190 5G modem (M.2 custom slot); Kill switch relay array; Nuclear security rail power management
Layer 4 — Power / Storage	Custom graphene-silicon primary cell (pouch); BV100 nuclear cell (isolated ground plane); QD solar MPPT controller; USB-C PD 100W controller; NVMe SSD (custom M.2 2230 format)
Rear Panel Assembly	Perovskite QD conformal solar film; protective Gorilla Glass 7i overlay; antenna array (5G × 4, WiFi × 2, UWB × 1)

8.3 Camera System

PARAMETER	SPECIFICATION
Primary Camera	50 MP, f/1.7, OIS, PDAF — standard wide
Ultrawide Camera	50 MP, f/2.2, 120° FOV

Telephoto Camera	12 MP, f/2.8, 5× optical zoom, OIS
Front Camera	12 MP, f/2.2 — under-display or notch-free punch-hole
ISP	RK3588 dual-core ISP; supplemented by QWAMOS ProVisual Engine (software)
Privacy	Hardware kill switch cuts camera power rail completely. Software: no camera API accessible to Guest VMs without explicit Dom0 policy grant.
Supplier	Sony Semiconductor (IMX series) or Samsung Semiconductor (ISOCELL) — per camera module

8.4 Audio

PARAMETER	SPECIFICATION
Microphone	Dual-microphone array (top + bottom) with hardware kill switch shorting lines to ground
Speaker	Stereo speakers — bottom-facing primary + earpiece secondary; Dolby Atmos certification target
Audio Codec	Qualcomm WCD series or AKM AK4377 — high-fidelity codec isolated from SoC audio path
ZRTP Integration	Audio path routed through QWAMOS ZRTP client for encrypted VoIP — hardware-accelerated in RK3588 DSP

SECTION 9 POST-QUANTUM CRYPTOGRAPHIC STACK

Full-Stack PQC — All Layers — No ECC Anywhere

9.1 Design Principle

VALKYRJA's cryptographic stack is designed around a single principle: **no ECC anywhere in the security-critical path.** ECDSA and X25519 are retained only as one component within hybrid constructions where classical algorithms are supplemented — never replaced — by post-quantum algorithms. The hybrid construction ensures that breaking any individual algorithm (whether by quantum computer or algebraic attack) does not compromise the overall security. Failure of all PQC algorithms simultaneously is required to compromise key material.

9.2 Key Encapsulation Mechanisms (KEMs)

Algorithm	Standard	Security Level	Role
ML-KEM-1024 (Kyber-1024)	NIST FIPS 203	Level 5 (AES-256 equivalent)	Primary KEM — all key exchanges
BIKE (Level 5)	NIST Round 4 Alternate	Level 5	Hybrid KEM component
HQC (Level 5)	NIST Round 4 Alternate	Level 5	Hybrid KEM component
Classic McEliece (Level 5)	NIST Round 4 Alternate	Level 5	Hybrid KEM component
X25519	RFC 7748	Classical 128-bit	Classical component in hybrid — not standalone
Hybrid Construction	Proprietary QWAMOS spec	Strongest of all components	KEM = ML-KEM XOR BIKE XOR HQC XOR McEliece XOR X25519

9.3 Digital Signature Schemes

Algorithm	Standard	Application
ML-DSA-87 (Dilithium)	NIST FIPS 204	Primary signing — hardware-backed in HSM; storage key signatures; firmware attestation
Falcon-1024	NIST FIPS 206	Compact signatures — certificate-constrained environments
SPHINCS+-SHA2-256	NIST FIPS 205	Stateless hash-based backup — immune to all algebraic attacks

9.4 Symmetric Primitives

Algorithm	Role	Quantum Status
ChaCha20-Poly1305	Authenticated encryption (AEAD) — all transport and storage encryption	Classical-secure (256-bit key → quantum-safe with current attacks)
BLAKE3	Hashing — integrity verification, key derivation input	Quantum-resistant at 256-bit output
Argon2id	Key derivation — GPU/ASIC-resistant	Memory-hard; resistant to dedicated

	KDF from passphrase	hardware attacks
HKDF-BLAKE2b	Key expansion — derives per-purpose keys from master key material	Classical-secure; quantum-safe at 256-bit output

9.5 QKD Simulation Protocols

QWAMOS includes software simulation of Quantum Key Distribution protocols for research, training, and future integration testing. These are not hardware QKD — they simulate the protocol logic in software pending hardware QKD infrastructure accessibility.

Protocol	Implementation	Status
BB84	crypto/pqc_next_gen/qkd_simulator.py	Functional simulation — Bennett-Brassard 1984 original protocol
E91	crypto/pqc_next_gen/qkd_simulator.py	Functional simulation — Ekert entanglement-based protocol
Decoy State	crypto/pqc_next_gen/qkd_simulator.py	Functional simulation — practical fiber-optic QKD defense model

SECTION 10 QWAMOS SOFTWARE SPECIFICATION

Operating System — v3.1.0 — 27/27 Phases Complete

10.1 Overview

PARAMETER	SPECIFICATION
Full Name	Qubes+Whonix Advanced Mobile Operating System
Version	3.1.0 (January 2026) — v3.2.0 targeted for device launch
License	AGPL-3.0 (open-source, copyleft — all derivatives must be open-source)
Repository	github.com/Dezirae-Stark/QWAMOS
Development Phases	27 / 27 complete (100%)
Codebase Size	50,000+ lines of code
Primary Languages	Python 3.9+, Dart/Flutter, TypeScript, Java/Kotlin
Architecture	ARM64-native KVM hypervisor OS with Qubes-style VM compartmentalization
Build System	Reproducible builds (repro-build.sh); SLSA Level 3 provenance; GPG-signed releases
Test Suite	Unit, integration, KVM hardware, differential KVM/QEMU, security, and hypervisor suites

10.2 Virtual Machine Domain Architecture

VM Domain	Network Access	Base OS	Purpose
Dom0 / Control VM	None	Minimal Linux	VM lifecycle, security policies, ML threat detection — privileged, no network
Gateway VM	Direct (filtered)	Whonix-style Linux	Tor/I2P/VPN/DNSCrypt routing — all user VMs route through here
Android VM	Via Gateway	AOSP Android 14+	Primary daily use environment — consumer app ecosystem
Arch Linux VM	Via Gateway	Arch Linux	Secure comms — Signal, ZRTP, steganography, PQC operations
Kali NetHunter VM	Via Gateway	Kali Linux	Offensive security, penetration testing toolkit
Ubuntu Dev VM	Via Gateway	Ubuntu 24.04 LTS	Development environment, AI App Builder pipeline
Vault VM	None (air-gapped)	Minimal Linux	Key storage, classified material, cryptographic operations
Disposable VM	Via Gateway	Ephemeral Debian	One-time-use — fresh per session, automatic wipe on close

10.3 User Experience — Profile Abstraction Layer

The QWAMOS hypervisor complexity is completely abstracted from the user through the Profile system. Users interact with four clearly branded Security Profiles. The underlying VM configuration, network routing, cryptographic state, and display mode are managed automatically by the profile engine in Dom0.

Profile	Display Name	VM Configuration	FMP State
Everyday	🏠 Everyday	Android VM active; Gateway VM routing; standard Tor circuit	Off
Private	🔒 Private	Fresh isolated Android VM; separate Tor identity; no cross-VM data	Standard Privacy
Secure	🛡️ Secure	Arch Linux VM; PQC comms; ZRTP voice; full E2E stack	Maximum Privacy
Ghost	👻 Ghost	Fresh Disposable VM; WebTunnel DPI defeat; Phantom Activity overlay	Maximum Privacy

10.4 AI & Machine Learning Stack

Component	Technology	Function
On-device LLM (Kali GPT)	Local inference via RK3588 NPU (6 TOPS)	Security analysis, threat assessment, exploit evaluation
External AI (Claude)	API via Tor (isolated channel)	Architecture design, code review, compliance guidance
External AI (ChatGPT)	API via Tor (isolated channel)	Code quality, UI/UX assistance, test generation
Consensus Engine	ai_security/orchestrator.py	Adjudicates conflicts between AI models; majority-rule synthesis
Threat Detection — Autoencoder	Unsupervised neural network (Dom0)	Novel/zero-day behavioral anomaly detection
Threat Detection — Random Forest	Supervised ensemble (Dom0)	Known attack signature and pattern matching
Threat Detection — LSTM	Recurrent neural network (Dom0)	Temporal sequence attack detection across time windows
AI Governor	hypervisor/ai_governor.py	ML-driven CPU/memory/I/O resource allocation per VM
LoRA Fine-tuning	ai_security/lora_finetuning.py	On-device model fine-tuning for threat intelligence adaptation

10.5 SecureType Keyboard

FEATURE	IMPLEMENTATION
Implementation	TypeScript / React Native — custom software keyboard
Per-Keystroke PQC Encryption	ML-KEM-1024 encryption applied to each individual keypress before leaving keyboard subsystem
Coordinate Obfuscation	±5 px random noise added to touch coordinates — defeats stylometric fingerprinting
Layout Randomization	Keyboard layout reshuffled every 30 seconds — defeats layout memorization shoulder surfing

Anti-Screenshot	FLAG_SECURE active whenever keyboard is visible — no screen capture possible
Clipboard Isolation	Clipboard contents encrypted and scoped per-VM — no cross-VM clipboard access
AI Typing Verification	Behavioral biometric model detects typing rhythm anomalies — alerts on unauthorized user

10.6 Software Required Modifications for VALKYRJA

The following QWAMOS components require modification or new development to support VALKYRJA hardware. These items are the primary software integration workstreams for the device launch:

- RK3588 hardware abstraction layer (HAL) — kernel drivers for RK3588 peripherals, camera ISP, audio codec, and power management within QWAMOS Linux kernel (6.6 LTS base)
- Display controller integration — QWAMOS profile state → FMP hardware signal path; display controller driver for Samsung LEAD 2.0 panel with FMP mode switching API
- HNCP firmware — RISC-V microcontroller firmware for Hardware Network Co-Processor; implements traffic isolation enforcement, Gateway VM watchdog, and kill switch protocol
- Tamper MCU firmware — Cortex-M0+ firmware for tamper detection microcontroller; implements tamper condition classification, alert routing, and HSM key destruction trigger
- HSM integration — Infineon SLB9 driver and key management API integration with QWAMOS PQC keystore; ML-DSA-87 hardware-backed operations
- Nuclear power rail management — Isolated power domain monitoring; HNCP and Tamper MCU power state management under nuclear cell
- Solar MPPT integration — Perovskite QD solar controller driver; charge path management to primary cell without cross-contaminating nuclear rail
- VM allocation tuning — Increase Dom0 and Guest VM memory ceilings to utilize full 32 GB LPDDR5X; update AI Governor CPU allocation policies for RK3588 8-core topology
- UFS 4.0 + NVMe dual storage — Storage controller driver update; Vault VM assigned to NVMe volume; per-VM encrypted volume key management update for dual-medium topology

SECTION 11 SUPPLIER PROCUREMENT TARGETS

Component Sourcing Strategy & Primary Contacts

11.1 Critical Path Components

The following components are on the critical procurement path. Supplier engagement must begin in parallel, not sequentially. Lead times for custom components (nuclear cell, graphene-silicon primary cell, FMP display panel) are 12–24 months for first production articles.

Component	Target Supplier(s)	Priority	Lead Time Est.
RK3588 SoC	Rockchip Electronics (direct); distributors: Arrow, Avnet, Mouser	P0 — Critical	8–16 weeks (stock); 16–24 weeks (custom config)
Samsung FMP LEAD 2.0 Display	Samsung Display Corporation — Mobile Display Business Division (B2B direct)	P0 — Critical	12–18 months (custom panel program)
32 GB LPDDR5X RAM	Samsung Semiconductor; SK Hynix; Micron Technology	P0 — Critical	12–16 weeks
1 TB UFS 4.0 Storage	Samsung Semiconductor (KLUDG8); SK Hynix (H28U1); Micron (MTFC)	P0 — Critical	12–16 weeks
Graphene-Silicon Primary Cell	Lyten Inc. (US); Graphene Manufacturing Group (AU); Sila Nanotechnologies	P0 — Critical	18–24 months (custom cell program)
Betavolt BV100 Nuclear Cell	Betavolt New Energy Technology Co. (Beijing) — direct engagement; NRC licensing parallel track	P0 — Critical	12–18 months + regulatory
Infineon HSM (SLB9672)	Infineon Technologies AG — Semiconductor Business Division; Digi-Key, Mouser	P1 — High	16–20 weeks
Sierra Wireless EM9190 5G Modem	Semtech Corporation (Sierra Wireless parent); Arrow Electronics	P1 — High	12–16 weeks
Perovskite QD Solar Film	Saule Technologies (PL); Quantum Solutions (Saudi/UAE); Greatcell Solar	P1 — High	18–24 months (custom deposition)
Hardware TRNG IC	Maxim Integrated (DS3900); STMicroelectronics (STSAFE-TPM); Microchip (ATAES132A)	P1 — High	8–12 weeks
WiFi 7 / BT 5.4 / UWB Module	Qualcomm FastConnect 7900; MediaTek Filogic 880; NXP SR150 (UWB)	P2 — Standard	12–16 weeks
Camera Modules	Sony Semiconductor (IMX989/IMX766); Samsung ISOCELL (HP9/GN5)	P2 — Standard	16–20 weeks
NVMe SSD (512 GB, custom M.2)	Samsung (PM9A3/custom); SK Hynix (PC801); Western Digital (SN850X custom)	P2 — Standard	12–16 weeks
Biometric Enclave IC	Synaptics (DSX100); Goodix (GH3x2x series); Egis Technology	P2 — Standard	12–16 weeks

11.2 Manufacturing Engagement Strategy

Device manufacturing should target ODM (Original Design Manufacturer) partners with experience in security device production and willingness to execute under strict IP protection agreements. Primary targets:

ODM Partner	Profile	Engagement Priority
Foxconn Technology Group (Hon Hai)	World's largest contract electronics manufacturer; significant security device experience; US and Taiwan facilities for ITAR-adjacent work	Primary
Pegatron Corporation	Apple ODM partner; high-quality assembly; significant security practices; Taiwan and Mexico facilities	Primary
Compal Electronics	Established security device experience; PCB integration capabilities; Taiwan	Secondary
Flextronics (Flex Ltd.)	US manufacturing facilities available; security clearance-adjacent capability; ITAR experience	Secondary (Tier 3 option)
Benchmark Electronics	US domestic manufacturing; specialized high-reliability electronics; suited for Tier 3 operational variant	Tier 3 Dedicated

SECTION 12 REGULATORY & COMPLIANCE FRAMEWORK

Certification Requirements & Jurisdiction Considerations

12.1 Required Certifications — All Tiers

CERTIFICATION	REQUIREMENT
FCC Part 15 (US)	Required for all RF-emitting devices sold in US market. Covers 5G modem, WiFi 7, Bluetooth, UWB, NFC. Testing at accredited laboratory (UL, Intertek, Bureau Veritas).
CE Mark (EU)	Required for EU market. Covers Radio Equipment Directive (RED), EMC Directive, RoHS. Notified Body engagement for RED certification.
PTCRB / GCF	Carrier certification for 5G/4G cellular bands. PTCRB for North America; GCF for international carriers. Testing per 3GPP TS specifications.
IC (Industry Canada)	Required for Canadian market. Coordinate with FCC process where possible.
IEC 62133 / UN 38.3	Lithium battery safety. Required for all custom graphene-silicon cell configurations. Testing at cell and pack level.
UL 2054	Household and commercial batteries — required for US market lithium cell certification.
RoHS 3 Compliance	Restriction of Hazardous Substances — affects component material selection across all PCBs.
WEEE Compliance	EU waste electrical and electronic equipment — registered producer requirement.
Bluetooth SIG Qualification	Required for use of Bluetooth trademark and royalty-free coexistence stack.
Wi-Fi Alliance Certification	Required for Wi-Fi 7 (802.11be) trademark use and interoperability certification.

12.2 Nuclear Cell — Regulatory Considerations

RADIOACTIVE MATERIAL LICENSING

Incorporation of Nickel-63 (Ni-63) radioactive isotope in the BV100 nuclear security cell requires engagement with the Nuclear Regulatory Commission (NRC) in the United States, or the national regulatory equivalent in the manufacturing jurisdiction. Specific requirements include:

- NRC License: Specific or general license for possession of byproduct material (10 CFR Part 30)
- Material quantity: BV100 contains approximately 20 curies Ni-63 per cell — above general license threshold
- Shipping: IATA Dangerous Goods Regulations (DGR) Section 10 — radioactive material; special packaging required
- End-of-life: Ni-63 decays to stable Cu-63 — no special disposal required at product end-of-life
- International: IAEA Safety Series No. 115 for transport; national nuclear authority licensing per jurisdiction

RECOMMENDATION: Engage NRC pre-application consultation concurrent with Betavolt procurement discussions.
Tier 3 Operational variant regulatory path may differ from Tier 1/2 consumer product path.

12.3 Export Control

FRAMEWORK	APPLICABILITY

EAR (Export Administration Regulations)	Post-quantum cryptography may be subject to EAR controls under ECCN 5E002 (software for military cryptography). Legal review required for Tier 3 operational export.
ITAR (International Traffic in Arms Regulations)	Tier 3 Operational variant targeting cleared personnel may trigger ITAR if device is classified as 'defense article.' Legal review required before Tier 3 specification is finalized for export.
Wassenaar Arrangement	Multi-lateral export control framework for dual-use technologies. Post-quantum cryptographic components require review in each export destination jurisdiction.

12.4 QWAMOS Software Compliance (Open Source)

ITEM	DETAIL
License	AGPL-3.0 — all derivatives and modifications must be released under same license if distributed
Contributor License Agreement (CLA)	Required from all contributors before code merge — assigns IP rights for potential license enforcement
Security Audit	Independent third-party security audit of QWAMOS codebase required prior to Tier 2/3 deployment. Recommended auditors: Trail of Bits, NCC Group, Quarkslab.
SLSA Provenance	Supply-chain level SLSA Level 3 provenance attestation for all release artifacts — reproducible builds with signed manifests
SBOM	Software Bill of Materials generated at each release via generate-sbom.sh — required for US government procurement under Executive Order 14028
Common Criteria	CC Target of Evaluation (TOE) framework implemented in qwamos_compliance/common_criteria.py — formal CC evaluation recommended for Tier 3

SECTION 13 CONSOLIDATED FULL DEVICE SPECIFICATION

Complete Technical Reference — Single Sheet

CATEGORY	PARAMETER	SPECIFICATION	NOTES
Compute	SoC	Rockchip RK3588	EL2/KVM confirmed — non-substitutable
Compute	CPU	4× Cortex-A76 @ 2.4 GHz + 4× Cortex-A55 @ 1.8 GHz	ARM DynamIQ big.LITTLE, ARMv8.2
Compute	GPU	Mali-G610 MP4	Vulkan 1.2, OpenGL ES 3.2, OpenCL 2.2
Compute	NPU	6 TOPS — 3-core neural processor	On-device AI inference
Compute	Process Node	8nm FinFET	Samsung Foundry
Memory	RAM	32 GB LPDDR5X	Maximum RK3588 config; 68.3 GB/s bandwidth
Memory	RAM Standard	JEDEC JESD209-5B	0.5V VDDQ ultra-low power
Storage	Primary	1 TB UFS 4.0	4,200 MB/s read; 2,800 MB/s write
Storage	Secondary	512 GB NVMe PCIe 3.0 x4	Vault VM dedicated; custom M.2 format
Storage	Total Internal	~1.5 TB	UFS + NVMe combined
Display	Panel	Samsung LEAD 2.0 AMOLED + Flex Magic Pixel	Privacy display — hardware FMP
Display	Size	6.9 inches	WQHD+ 3088 × 1440
Display	Refresh Rate	1–120 Hz LTPO 4.0 adaptive	
Display	Peak Brightness	2,600 nits	HDR10+
Display	Privacy (45°)	3.5% front brightness retained	vs. ~40% standard OLED
Display	Privacy (60°)	0.9% or less	Screen appears black to bystanders
Power	Primary Cell	Graphene-silicon Li-ion, 7,700 mAh eff., ~35 Wh	+40% vs. standard Li-ion + QD electrode
Power	Charge Time	6–10 minutes (USB-C PD 100W)	
Power	Solar Harvesting	Perovskite QD conformal film, 50–150 mW ambient	Rear panel — trickle charge
Power	Nuclear Security Cell	Betavolt BV100, 100 µW, 50-year life	Dedicated security rail — isolated

Security	HSM	Infineon SLB9672 TPM 2.0 class	PQC key management, attestation
Security	TRNG	Hardware true RNG — ring oscillator + shot noise	All crypto seeded from hardware entropy
Security	HNCP	RISC-V MCU <40 µW — on-wire modem gating	Nuclear-powered; always-armed
Security	Tamper MCU	ARM Cortex-M0+ <1 µW — physical tamper detection	Nuclear-powered; zero-battery operation
Security	Biometric Enclave	Discrete ARM SecurCore SC300	Fingerprint isolated from SoC
Security	Kill Switches	4× physical relay — Network, Mic, Camera, Location	Opens circuits — no software override
Connectivity	5G	Sub-6 GHz + mmWave	Sierra Wireless EM9190 isolated M.2
Connectivity	WiFi	WiFi 7 (802.11be) tri-band	2.4 / 5 / 6 GHz
Connectivity	Bluetooth	5.4	LE Audio, LC3
Connectivity	UWB	IEEE 802.15.4z	Secure ranging, attestation
Connectivity	NFC	Type A/B/F, ISO 14443/15693	
Connectivity	USB	USB-C 3.1 Gen 1, PD 3.1 100W, DisplayPort Alt	
Camera	Primary	50 MP, f/1.7, OIS, PDAF	Sony IMX / Samsung ISOCELL
Camera	Ultrawide	50 MP, f/2.2, 120° FOV	
Camera	Telephoto	12 MP, f/2.8, 5× optical, OIS	
Camera	Front	12 MP, f/2.2	
Software	OS	QWAMOS v3.1.0+ (AGPL-3.0)	27/27 phases complete
Software	VMs	8 isolated KVM domains	Dom0, Gateway, Android, Arch, Kali, Ubuntu, Vault, Disposable
Software	Network	Mandatory Tor/I2P — no clearnet bypass	WebTunnel DPI defeat; behavioral obfuscation
Software	PQC	Full stack — ML-KEM-1024, ML-DSA-87, ChaCha20	Hybrid constructions; no standalone ECC
Software	UI	Flutter (Dart) — 4 Security Profiles	Material Design 3; hypervisor complexity abstracted

Physical	Chassis	CNC aluminum unibody, IP68	6000-series alloy
Physical	Dimensions (est.)	163 × 78 × 9.2 mm	Subject to thermal validation
Physical	Weight (est.)	245–265 g	
Physical	Rear Panel	Gorilla Glass 7i + QD solar conformal film	

APPENDIX A — BILL OF MATERIALS SUMMARY

The following table provides a high-level component Bill of Materials for procurement planning. Unit costs are indicative estimates for small-volume (1,000–10,000 unit) production and are subject to supplier negotiation, volume discounts, and market conditions. Costs in USD.

Component	Supplier Target	Est. Unit Cost (Small Vol.)	Notes
RK3588 SoC	Rockchip / Arrow / Avnet	$35–55	Volume pricing at 10k+ units significantly lower
32 GB LPDDR5X RAM	Samsung / SK Hynix	$45–65	PoP integration adds $5–10 for packaging
1 TB UFS 4.0 Storage	Samsung / SK Hynix	$40–60	
512 GB NVMe SSD	Samsung / SK Hynix / WD	$25–40	Custom M.2 form factor adds tooling cost
Samsung FMP LEAD 2.0 Display	Samsung Display Corp.	$80–130	Custom panel; pricing TBD via direct B2B; MOQ critical
Graphene-Silicon Primary Cell	Lyten / GMG / Sila	$60–120	Custom cell; early-stage supplier; high uncertainty
Betavolt BV100 Nuclear Cell	Betavolt (Beijing)	$200–500	Highly uncertain; regulatory cost overhead not included
Perovskite QD Solar Film	Saule / Quantum Sol.	$30–60	Custom conformal deposition; area-dependent
Infineon HSM (SLB9)	Infineon / Digi-Key	$8–15	
Sierra Wireless EM9190 5G	Semtech / Arrow	$45–75	Includes carrier certification cost amortization
WiFi 7 / BT / UWB Module	Qualcomm / MediaTek	$12–20	
Camera Modules (3×)	Sony / Samsung ISOCELL	$25–45	Per camera module; 3 required
Biometric Enclave + Under-display Sensor	Synaptics / Goodix	$15–25	
Hardware TRNG	Maxim / STM / Microchip	$3–6	
HNCP RISC-V MCU	SiFive / RISC-V vendor	$2–5	Custom firmware development cost separate
Tamper Detection MCU	STMicroelectronics	$1–3	
NFC Controller	NXP / ST	$2–4	
USB-C PD Controller	Cypress / TI	$3–6	100W PD 3.1 capable
PCB (multilayer custom)	TTM / Tripod / Shennan	$15–30	6–8 layer custom stack
Chassis + Mechanical	Foxconn / Pegatron tooling	$25–50	Includes CNC aluminum unibody

			amortization
Miscellaneous passives / connectors	Various	$8–15	
ESTIMATED TOTAL BOM	—	$690–1,349	Excluding: regulatory, certification, software, tooling amortization

COST REDUCTION TRAJECTORY

BOM cost at small volumes (1,000–10,000 units) is elevated by custom component premiums, particularly the graphene-silicon cell, nuclear security cell, and FMP display panel. At 100,000+ unit scale, BOM cost is expected to fall to the $350–550 range as custom cell costs normalize and panel pricing improves through volume commitment. Tier 1 consumer device at high volume funds the cost structure for Tier 2/3 premium variants.

APPENDIX B — QWAMOS DEVELOPMENT PHASE COMPLETION

All 27 QWAMOS development phases are complete as of version 3.1.0 (January 2026). The following table records phase completion status for supplier and partner reference.

Phase	Name	Status	Key Deliverables
1	Core Hypervisor	Complete	ARM64 KVM hypervisor, VM lifecycle management
2	Network Isolation	Complete	Whonix-style gateway, Tor/I2P routing
3	Storage Encryption	Complete	Per-VM encrypted volumes, key management
4	Post-Quantum Crypto	Complete	ML-KEM-1024, ChaCha20-Poly1305, BLAKE3, Argon2id
5	Firewall & Policies	Complete	Per-VM firewalls, 12 policy toggles, panic wipe
6	AI Integration	Complete	Triple-AI coordination — Kali GPT, Claude, ChatGPT
7	ML Threat Detection	Complete	Autoencoder, Random Forest, LSTM detection models
8	SecureType Keyboard	Complete	Per-keystroke PQC, coordinate obfuscation, layout shuffle
9	AI App Builder	Complete	Triple-AI code generation, automated security audit
10	Hardware Security	Complete	ML bootloader lock, Vault 7 mitigation
11	Flutter UI System	Complete	26 custom widgets, GLSL shaders, Material Design 3
12	KVM Acceleration	Complete	VirtIO drivers, hardware virtualization optimization
13	PQC Storage	Complete	ML-DSA-87 hardware-backed keys, StrongBox integration
14	GPU Isolation	Complete	Per-VM GPU slices, Vulkan passthrough
15	AI Governor	Complete	ML resource allocation, predictive VM scaling
16	Cluster Mode	Complete	Multi-device mesh, live VM migration across devices
17	HSM Integration	Complete	TPM 2.0, FIDO2, TrustZone integration
18	Network Privacy (Enhanced)	Complete	WebTunnel, V2Ray, behavioral obfuscation (v3.1.0)
19	Offensive Security	Complete	Vulnerability scanner, penetration testing toolkit
20	Enhanced AI/LLM	Complete	Multi-model orchestration, LoRA fine-tuning
21	App Security	Complete	RASP, package verification, auditing
22	Secure Comms	Complete	Signal Protocol, ZRTP, steganography
23	Forensics	Complete	Memory forensics, chain of custody
24	Compliance	Complete	CIS, NIST 800-53, OWASP, Common Criteria
25	UX / Accessibility	Complete	React Native UI, voice control, gesture navigation
26	IoT Ecosystem	Complete	Device discovery, MQTT/Zigbee/Z-Wave, wearables
27	Quantum Resistance	Complete	BIKE, HQC, McEliece, QKD simulation, hybrid crypto

APPENDIX C — COMPETITIVE SECURITY ARCHITECTURE COMPARISON

This appendix documents VALKYRJA's security capabilities against the current competitive landscape for reference in manufacturer and supplier briefings.

Capability	VALKYRJA	S26 Ultra	Pixel 10 Pro	Librem 5	Bittium Tough
KVM / Type-1 Hypervisor	**Yes — native**	No	Partial (AVF)	Yes (slow)	No
VM Domain Isolation	**8 domains**	None	1 (Microdroid)	Yes	No
RAM	**32 GB**	12–16 GB	16 GB	4 GB	4 GB
Storage	**1.5 TB**	256 GB–1 TB	128 GB–1 TB	32 GB	64 GB
Post-Quantum Crypto (full stack)	**Yes**	No	Partial	No	Partial
Mandatory Tor for all traffic	**Yes**	No	No	No	No
Hardware Kill Switches (physical)	**4× relay**	None	None	3× switch	1× (network)
Discrete HSM	**Yes (Infineon)**	TrustZone	Titan M2	No	Yes
Privacy Display (FMP)	**Yes + FLAG_SECURE**	FMP only	None	None	None
Nuclear Security Rail	**Yes (BV100)**	None	None	None	None
Solar Harvesting	**Yes (QD perovskite)**	None	None	None	None
6-Minute Charge	**Yes (graphene-silicon)**	No (~80 min)	No (~70 min)	No	No
Anti-Surveillance (ultrasonic)	**Yes**	No	No	No	No
Behavioral Obfuscation (ML)	**Yes**	No	No	No	No
DPI-Defeating Transport	**WebTunnel**	None	None	None	None
Per-Keystroke PQC Encryption	**Yes (SecureType)**	No	No	No	No
Open Source OS	**Yes (AGPL-3.0)**	No	Partial	Yes	No
Consumer-Grade UX	**Yes**	Yes	Yes	No	No

[COMPANY NAME]

Technical Product Memorandum — [PRODUCT NAME] Codename: VALKYRJA

PROPRIETARY & CONFIDENTIAL — All Rights Reserved — April 2026

Prepared by Desirae Stark, Chief Architect

QWAMOS v3.1.0 — github.com/Dezirae-Stark/QWAMOS